A Multi-Stage Decision-Support Pipeline for Chest X-Ray Abnormality Classification, Localization, and Structured Reporting

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***Abstract*—Chest X-ray interpretation is a high-volume clinical task, where automated tools should provide more than a binary abnormality score. A useful decision-support system should combine image-level predictions, spatial evidence, and a clear explanation that a clinician can review. This project develops an explainable chest X-ray pipeline on a 15-label subset of the VinBigData chest X-ray dataset. The pipeline combines multi-label classification, YOLO-based localization, and an agentic reporting layer that converts model outputs into structured summaries. We compare a CNN trained from scratch with ImageNet-pretrained EfficientNet, ResNet, Vision Transformer, and Swin Transformer models. The best image-level classifier is Swin Base Patch4 Window7 224, which reaches a macro AUC-ROC of 0.9618. A simple CNN trained from scratch remains competitive at 0.9080, while a non-pretrained ViT model performs worse at 0.8417. For localization, YOLOv8m gives a 0.3418 mAP@0.5 after 100 epochs of fine tuning. Overall, this work presents a multi-stage decision-support pipeline that combines strong classification, useful localization evidence, and standardized reporting.**

***Index Terms*—Chest X-ray, VinBigData, medical imaging, multi-label classification, Swin Transformer, YOLO, explainable AI, radiology report generation, decision support.**

I. Introduction

Chest radiography is one of the most common imaging studies in clinical medicine. Because of its scale and importance, it is a strong setting for decision-support tools that can help clinicians review images more efficiently and consistently. However, a model that only returns class probabilities is hard to use in practice. Clinicians often need to know not only *what* was predicted, but also *where* the finding appears, how strongly the system supports it, and whether the different parts of the pipeline tell a consistent story.

This project addresses that need by treating chest X-ray analysis as more than a classification task. Our goal is to build a practical decision-support pipeline that combines image-level classification, localization, and structured report-style explanation. Given a chest X-ray, the system predicts abnormality labels, highlights candidate regions with bounding boxes, and produces a standardized summary that is easier to review.

The main question of this report is:

> Can a chest X-ray pipeline combine strong image-level classification, useful localization evidence, and structured reporting into a clearer and more interpretable decision-support workflow?

Our results suggest that it can. Swin Transformer provides the strongest image-level classification performance, while YOLO adds useful localization evidence, even though there is still room to improve detection quality. The reporting layer helps present the outputs in a more organized and readable form. Taken together, these components support a positive step toward a more interpretable and practical chest X-ray decision-support system.

II. Problem Setup and Dataset

This work uses a 15-label subset of the VinBigData chest X-ray dataset available in the current repository. The labels are: Aortic enlargement, Atelectasis, Calcification, Cardiomegaly, Consolidation, ILD, Infiltration, Lung Opacity, Nodule/Mass, Other lesion, Pleural effusion, Pleural thickening, Pneumothorax, Pulmonary fibrosis, and No finding.

The task is multi-label image classification because a single chest X-ray can contain multiple findings. In addition, the original bounding box annotations are used to train a localization model. The classification and detection tasks are connected downstream through the agentic reporting layer, but they are evaluated separately because they answer different clinical questions. Classification asks whether an abnormality is likely present in the image. Localization asks whether the system can identify a plausible spatial region for the finding.

III. Pipeline Overview

The final workflow is:

$$\text{CXR image} \rightarrow \text{Swin classifier} \rightarrow \text{YOLO localizer} \rightarrow \text{agentic reviewer} \rightarrow \quad (1)$$

The agentic reviewer does not replace the trained vision models. It consumes their outputs, including class probabilities, predicted labels, YOLO detections, bounding boxes, and agreement or disagreement between classification and detection. It then generates a structured decision-support summary containing supported findings, uncertain findings, confidence bands, a short impression, review recommendations, and a safety disclaimer.

IV. Methodology

A. Image-Level Classification

The classification pipeline was simplified to a single reproducible multi-label setup. All models were trained using 512×512 resized inputs, fixed train-validation splits, and binary cross-entropy with logits:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{NC}\sum_{i=1}^{N}\sum_{c=1}^{C}\left[y_{ic}\log\sigma(z_{ic}) + (1-y_{ic})\log(1-\sigma(z_{ic}))\right], \tag{2}$$

where N is the number of images, $C = 15$ is the number of labels, y_{ic} is the binary ground-truth label for class c, and z_{ic} is the model logit.

The retained classification models were a from-scratch CNN baseline, EfficientNet-B3, ResNet-50, ViT-Base, and Swin Base Patch4 Window7 224. The transfer-learning models used ImageNet-family pretrained weights where applicable. This comparison was designed to separate the benefit of architecture scale from the benefit of pretraining.

B. Localization with YOLOv8m

The localization branch was trained using YOLOv8m on the raw VinBigData box annotations. YOLO was evaluated in two ways. First, it was evaluated as an object detector using mAP@0.5 and mAP@0.5:0.95. Second, it was evaluated as a derived image-level predictor by converting detections into label presence or absence. This second view is not meant to replace the classifier evaluation; it measures how much classification signal is recoverable from localized detections alone.

C. Agentic Reporting Layer

The reporting layer maps model outputs into a standardized report. Its inputs include the full Swin probability vector, thresholded predicted labels, YOLO class detections, bounding boxes, and agreement or disagreement between the classifier and detector. The output includes supported findings, uncertain findings, confidence bands, a short impression, review recommendation, and safety disclaimer.

The reporting layer also maps low-level labels into derived global buckets for readability:

- No acute abnormality
- Cardiomediastinal abnormality
- Pleural abnormality
- Airspace or infectious-inflammatory pattern
- Chronic interstitial or fibrotic pattern
- Focal lesion or mass-like pattern
- Possible urgent thoracic abnormality

These buckets are interpretive summaries, not additional ground-truth labels.

TABLE I
Image-Level Classification Results

Model	Type	Params	Macro AUC-ROC
simple_cnn	Scratch CNN	1.31M	0.9280
efficientnet_b3	Transfer	11.49M	0.9504
resnet50	Transfer	24.56M	0.9453
vit_base	Transformer	86.84M	0.8417
swin_base	Hierarchical Transformer	87.28M	0.9618

TABLE II
YOLOv8m Localization and Derived Image-Level Results

Run	mAP@0.5	mAP@0.5:0.95	Exact	Macro Acc	Macro F1
10 epochs	0.1723	0.0981	0.7470	0.9501	0.3188
50 epochs	0.2452	0.1295	0.7473	0.9567	0.5422

V. Experimental Setup

All reproduced classification runs were executed on the `oc4` compute environment across four H100 GPUs. The from-scratch CNN was trained for 100 epochs. Transfer-learning models were trained for 10 epochs. YOLO was first trained for 10 epochs and later improved using a 50-epoch four-GPU run.

The primary image-level classification metric is macro AUC-ROC. Macro averaging is appropriate because the project uses multiple labels and some abnormalities may be less frequent than others. For YOLO, the primary detection metrics are mAP@0.5 and mAP@0.5:0.95. Derived image-level YOLO predictions are additionally summarized using exact-match accuracy, macro accuracy, and macro F1.

VI. Classification Results

Table I summarizes the image-level classification results. Swin Base Patch4 Window7 224 achieves the best macro AUC-ROC at 0.9618. EfficientNet-B3 and ResNet-50 are also strong, with macro AUC-ROC values of 0.9504 and 0.9453 respectively. The simple CNN baseline reaches 0.9280, which is surprisingly competitive given its much smaller parameter count. ViT-Base underperforms in this setup, reaching 0.8417.

The main conclusion is that ImageNet-family pretraining helps substantially, but model architecture also matters. Swin's hierarchical design appears better suited to this setup than the ViT path used here. At the same time, the simple CNN baseline is valuable because it demonstrates that the classification problem is learnable even without a very large transformer.

VII. YOLO Localization Results

Table II shows the YOLO localization results. The initial 10-epoch run obtains 0.1723 mAP@0.5 and 0.0981 mAP@0.5:0.95. After extending training to 50 epochs using four GPUs, the detector improves to 0.2452 mAP@0.5 and 0.1295 mAP@0.5:0.95. Derived image-level macro F1 improves from 0.3188 to 0.5422.

The improved YOLO run confirms that longer training and better compute utilization materially improve localization. However, YOLO does not surpass Swin as the primary image-level predictor. Its strongest role is as a supporting localization branch that provides spatial evidence for the reporting layer.

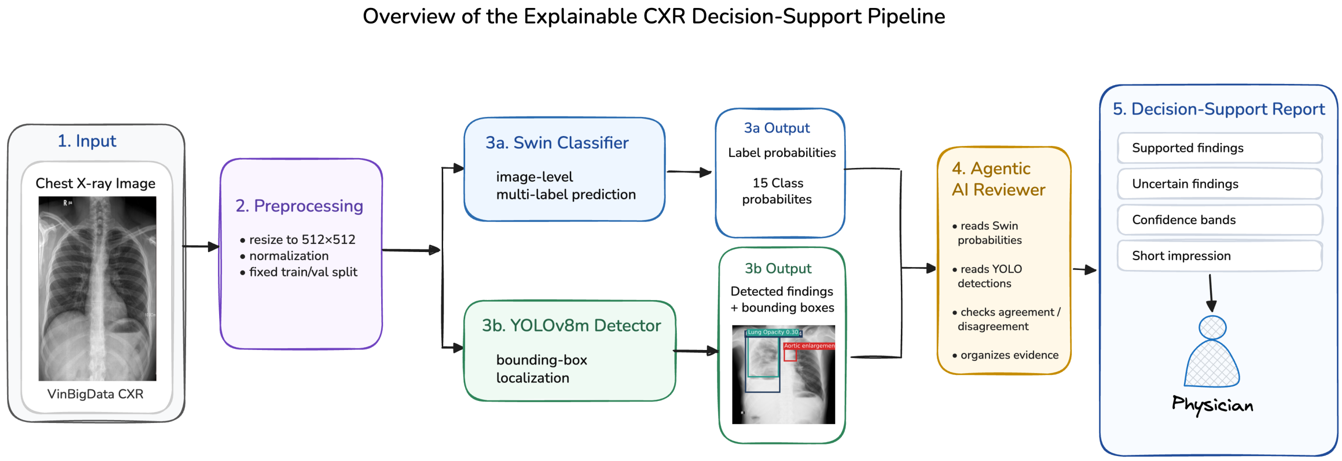

Fig. 1. Overview of the proposed explainable chest X-ray decision-support pipeline. The workflow combines preprocessing, image-level classification with Swin, localization with YOLOv8m, and an agentic reviewer that organizes the outputs into a structured report for clinical review.

Placeholder for Fig. 2: Classification Bar Chart

Plot macro AUC-ROC for simple CNN, EfficientNet-B3, ResNet-50, ViT-Base, and Swin.

Fig. 2. Classification performance comparison. Replace this placeholder with a grouped or single-metric bar chart showing macro AUC-ROC.

Placeholder for Fig. 3: YOLO Examples

Show original CXR with ground-truth boxes and YOLO predicted boxes.

Fig. 3. Example localization output. Replace this placeholder with qualitative YOLO predictions comparing ground-truth boxes and predicted boxes.

VIII. Structured Reporting Results

The final system generates presentation-quality reports that compare the original image, ground-truth radiologist labels, Swin predictions, YOLO predictions, YOLO bounding boxes, and a structured agentic summary. The reporting format is intentionally standardized to avoid generic language model prose. The goal is to make the model output easier to audit by exposing the evidence used by the system.

The agentic layer is best interpreted as an explainability and communication layer. It does not improve raw macro AUC-ROC directly, but it improves the usability of the pipeline by converting raw probabilities and boxes into a form that resembles clinical reasoning.

IX. Discussion

The strongest result is the Swin classifier's macro AUC-ROC of 0.9618. This suggests that hierarchical transformer features are effective for this chest X-ray classification setup. EfficientNet-B3 and ResNet-50 also perform strongly, supporting the value of transfer learning from natural-image pretraining.

The simple CNN result is also important. With only 1.31M parameters, it reaches a macro AUC-ROC of 0.9280. This makes it a useful sanity-check baseline and prevents the project from overstating the necessity of large transformers. A report that only included Swin would be less convincing because it would not show whether the gain came from the architecture, pretraining, or the task being relatively learnable.

The YOLO branch provides a different kind of value. Its mAP values remain modest, but the 50-epoch run improves substantially over the initial run. In a clinical decision-support pipeline, localization does not need to replace classification to be useful. Even imperfect boxes can help make the prediction more reviewable, especially when the report explicitly distinguishes supported findings from uncertain findings.

X. What the Numbers Tell Us

- Swin Base Patch4 Window7 224 is the strongest image-level classifier, with a macro AUC-ROC of 0.9618.
- EfficientNet-B3 and ResNet-50 are strong transfer-learning baselines, reaching 0.9504 and 0.9453 macro AUC-ROC respectively.
- The from-scratch CNN is a surprisingly strong lightweight baseline at 0.9280 macro AUC-ROC.
- ViT-Base underperforms in this setup at 0.8417 macro AUC-ROC, suggesting that transformer scale alone is not sufficient.
- YOLOv8m improves with longer training, increasing mAP@0.5 from 0.1723 to 0.2452 and derived macro F1 from 0.3188 to 0.5422.

Placeholder for Fig. 4: Example Generated Report

Include original image, ground-truth labels, Swin probabilities, YOLO boxes, supported findings, uncertain findings, and short impression.

Fig. 4. Example structured decision-support report. This figure should show one full report page or a compact report excerpt.

- The agentic reporting layer is useful as an interpretability and communication layer, but it should not be treated as a clinically validated reporting system without further evaluation.

XI. Limitations and Future Work

This project has several limitations. First, the report currently emphasizes aggregate macro AUC-ROC for classification. Per-class AUC, F1, sensitivity, specificity, and calibration would make the evaluation more clinically meaningful. Second, YOLO localization remains relatively weak, especially under the stricter mAP@0.5:0.95 metric. Third, the agentic reporting layer has not yet been evaluated by clinicians or scored against radiologist reports. Fourth, the current analysis does not include uncertainty calibration, threshold tuning, subgroup analysis, or external validation.

The most important future steps are:

- Add per-class classification metrics for all 15 labels.
- Add ROC and precision-recall curves for the strongest classifier.
- Evaluate calibration using reliability curves and expected calibration error.
- Report threshold selection strategy rather than relying only on default thresholds.
- Add qualitative success and failure cases for YOLO boxes.
- Evaluate the generated reports using a structured rubric: correctness, completeness, hallucination rate, and usefulness.
- Add a pipeline flow diagram and example report figure.

XII. Conclusion

This project successfully moves beyond a basic chest X-ray classification benchmark toward a more interpretable medical AI workflow. The strongest classifier is Swin, which reaches 0.9618 macro AUC-ROC on the 15-label VinBigData subset. YOLO adds localization support, improving meaningfully with longer multi-GPU training, though it remains weaker than the classifier as a primary predictor. The agentic layer converts raw predictions and detections into structured summaries that are easier to audit and present. The final system is therefore best understood as a multi-stage decision-support pipeline: classification provides predictive strength, localization provides spatial evidence, and structured reporting provides interpretability.

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The author thanks the course staff and project collaborators for feedback and compute support. Replace this sentence with the actual names or remove the section if acknowledgments are not needed.

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